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Assignment - 2

Q1. List and explain four basic activities in Interaction Design.

→ The four basic activities in Interaction Design are:

i) Establishing Requirements:

- Purpose: This involves understanding what the users need from the system or product. It's about identifying the task and challenges of users to ensure the design meets their needs.

- Explanation: Designers gather information about who the users are, what they need, and the environment in which they will use the product.

ii) Designing Alternatives:

- Purpose: This stage involves gathering a variety of design solutions to address the requirements gathered in first stage.

- Explanation: Designers create different prototypes or concepts, exploring multiple approaches to solving

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the design problem.

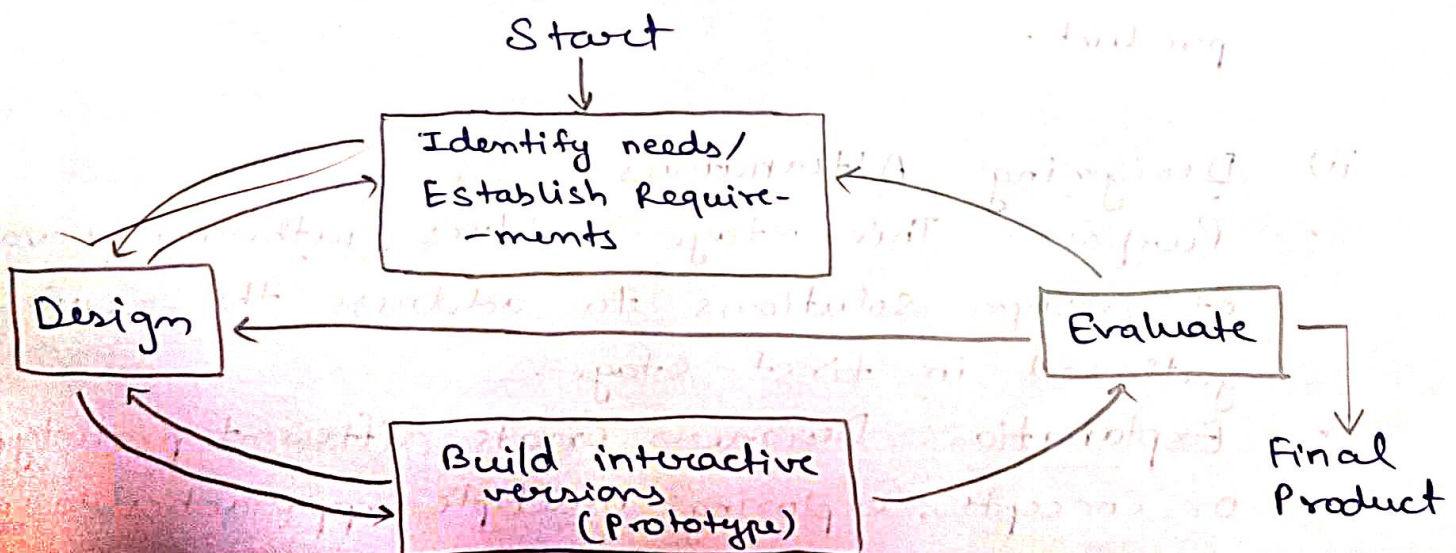
iii) Prototyping :

- Purpose : Prototyping is about create tangible representation of the design concepts to test and refine them.
- Explanation : Prototypes can range from low-~~fidelity~~ fidelity sketches to high-fidelity interactive models. They allow designers to explore how users interact with the design.

iv) Evaluating :

- Purpose : This involves assessing the effectiveness, efficiency and satisfaction of the design.
- Explanation : Through usability testing, user feedback and performance metrics, designers gather data on how well the design meets user needs and where improvement can be made.

Q2: Draw and Explain life cycle mode for interactive system.



i) Requirements (Establishing Needs) :-

Understand and define what the users need from the interactive system. In this phase, designers and developers gather information about the target users, their tasks & the environment in which the system will be used.

ii) Design Alternatives (Generating Solutions) :-

Create various possible solutions or design alternatives that can meet the requirements. Designers brainstorm and develop multiple design concepts. The goal is to explore different approaches to solving the design.

iii) Prototyping (Building Representation) :-

Develop tangible versions of the design alternatives to test their feasibility and usability. These prototypes allow designers to explore how the user interacts with the system.

iv) Evaluation (Testing and Refining) :-

Assess the prototypes to ensure they meet user needs and are usable. This phase helps in identifying usability problems and understanding user satisfaction with design.

v) Iteration :

Refine and improve the design based on the evaluation.

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result. This iterative cycle continues until the design is optimized and meets the desired user requirements.

vi) Final Product:

Deliver a well-designed interactive systems that satisfies user needs and is ready for development. The final product should be user friendly, efficient and aligned with the initial requirements.

Q3: Give meaning of 1.) Learnability, 2.) Flexibility, 3.) Robustness

1.) Learnability :-

It refers to how easy it is for users to learn and become proficient with a system or interface. A system with high learnability allows users to quickly grasp how to use it, often with minimal instruction or experience.

2.) Flexibility :-

It refers to the system's ability to adapt to different user needs and preferences. It means providing multiple ways for users to accomplish tasks or interact with the system, catering to a variety of skill.

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3.)

Robustness:-

It refers to the system's ability to support users in achieving their goals, even when unexpected situation arise.

It includes providing clear feedback, error recovery options and preventing user from making serious mistake.

Q4. Discuss MVC and PAC model with proper schematic.

→ Model-View-Controller (MVC) Architecture, the MVC is a software design pattern commonly used for developing user interfaces. It divides the application into 3 interconnected component:-

- Model:- Manages the data and business logic.

It represents the application's data and the rules that govern access to and updates of this data. The model is independent of user interface.

- View:- Represents the user interfaces and displays the data from the model to user. It receives input from the user and sends it to the controller for processing.

- Controller:- Acts as an intermediary b/w the model and the view. It takes the user's input

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from the view, processes it and update the view accordingly.

→ Presentation-Abstraction-Control (PAC) :-

The PAC is another architectural pattern similar to MVC but is more hierarchical and is used in distributed systems.

Presentation :- Corresponds to the view in MVC and manages the user interface. It provides a way for the users to interact with application.

Abstraction :- Handles the data and business logic, similar to the model in MVC. It is responsible for the data processing and stores the application state.

Control :- Acts as intermediary and handles the communication b/w the Presentation and Abstraction layers. It also manages the flow of control in the system.

Q5. List and explain various evaluation techniques for interaction designs.

→ Evaluating interactions design is crucial to ensure that they meet user needs and goal.

Usability Testing :-

Purpose :- To observe users interacting with the design and identify any usability issues.

Methods :-

→ In-Person testing :- User interact with the design in a controlled environment while being observed.

→ Remote testing :- Users interact with the design remotely, often using screen sharing software.

• Key Metrics :- User satisfaction
Error rates

ii) Heuristic Evaluation :-

Purpose :- To evaluate the design against a set of usability heuristics.

Method :- Experts evaluate the design using checklist of heuristics.

• Key Metrics :- Number of usability problems identified

iii) Cognitive Walkthrough :-

Purpose :- To evaluate the design from a user's perspective, step-by-step.

Methods :- → User's goals

→ Actions needed to achieve goals.

→ Whether the design provides clear cues and feedback.

• Key Metrics :- Number of usability problems identified.

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iv) User Surveys :-

- Purpose: To gather user feedback on the design.
- Methods: Distribute survey to users after they have interacted with design.
 - Ask questions about user satisfaction.
- Key Metric: User satisfaction ratings.

v) Expert Review :-

- Purpose: To get feedback from experts in the field.
- Method:
 - Invite experts to review the design and provide feedback.
 - Experts can offer insights based on their knowledge.
- Key Metrics: Expert feedback and recommendations.

Q6. Explain Persona and its importance in Interaction Design?

- A persona is a fictional character that represents a specific user type within a targeted demographic, attitude and behaviour set.
- ID, personas are to guide the design process by creating a more accurate representation of the end users. They help designers and developers understand the needs, experiences, behaviour and goals of the people they are designing for.



Importance of Personas in Interaction Design:

- i) User Centered Design :- Personas keep the design process focused on the users. By developing a deep understanding of who the users are, designers can create more intuitive and user-friendly interfaces.
- ii) Guided Decision - Making :- Personas provide a reference point for making design decisions. They help ensure that the final design aligns with the user's preference and behaviour.
- iii) Scope Management :- Personas help in managing the scope of the project by clearly defining who the primary users are.
- iv) Enhanced communication :- Personas make it easier to communicate user needs and design rationale to stakeholders, as they can visualize the users and understand their needs.

Imps